CA-1.82 - Discrete Mathematics

Chapter-1 Mathematical logic and proof

1.1 proposition: Proposition is (statement) is a sentence which has a truth value (either true or False) but not both

e.g 2 is even number = true = prop

Triangle has four sides = False = prop

Give me that book - not proposition

1.1.1 Logical Connectives

a) Conjuction (and) ($\wedge$ - symbol)

b) Disjunction (or) ($\vee$ - symbol)

c) Implication (~~If then~~ Implies, If -- then ---)

symbol ($\longrightarrow$)

d) bi-implication (If ... and only if ...)

symbol ($\longleftrightarrow$)

detail of c and d logical Connectives

c) $p \longrightarrow q$

1) p implies q
2) If p then q
3) q if p
4) p only if q
5) p is sufficient for q
6) q is necessary for p

d) $p \longleftrightarrow q$

1) p if and only if q
2) p implies q and q implies p.
3) p is necessary and sufficient for q
4) p is equivalent to q.

Examples ①

p: Addis A. is Capital City of Ethiopia

q: 123 is prime number

problem: Construct using logical Connection to get new proposition.

② If ~~the~~ it is morning, then the sun is in the east

a) write maths material representation

b) write converse, inverse, Contra

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1.3 Equivalent proposition.

Two Compound propositions, P and Q are said to be equivalent if they have the same truth value for all possible combination of truth values for the component proposition.

written $P \equiv Q$.

Example: $p: p \rightarrow q$

$Q: \neg q \rightarrow \neg p$

1.4 truth table and Tautology

Defn: A Compound proposition is a tautology if it is always true regardless of the truth values of its Component proposition.

Eg. $p \rightarrow (q \rightarrow p)$ for any proposition p and q is tautology

1.5 Arguments and proof of predicate logic

Proof of predicate logic

Defn: An open statement (also called predicate logic) is a sentence that contain one or more variables and whose truth values depend on the value we assigned for the variables.

Example: let $p(x)$: $x^2 - 1 = 0$

$U = \{-1, 1, 0, -1/2\}$

$p(x)$ = true or false for $x \in U$

$Q(x)$: $|x+1| > 1$

$U = \{-1, 1, 0, -1/2\}$ = universal set

Note: Both open proposition $p(x)$ and $Q(x)$ are said to be equivalent prop.

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Note 7

P(x) is tautology iff P(a) = true for all a.

- Quantifier

a) universal quantifier

b) existential quantifier

a) Universal Quantifier

- for every x, for all x, for each x ⟶ ∀x.

∀x P(x) read as every x has property P.

b) existential quantifier

- there exist an x, for some x for atleast one x ⟶ ∃x

(∃x)P(x) ⟶ there exists x∈U with property P.

Example: P(x): $x^2+x-6=0$

U = IRT = {x : x > 0}

∃x P(x) = true

∀x P(x) = false

Combination of Quantifier

1) (∀x)(∀y) P(x,y)

⟶ for all x and for all y the relation P(x,y) holds

2) (∃x)(∃y) P(x,y)

" there is x and there is y for which P(x,y) holds

3) (∀x)(∃y) P(x,y)

" for every x there is a y such that P(x,y) holds

here y depends on the value of x.

4) (∃x)(∀y) P(x,y)

" there exists an x which stands for all y in the relation P(x,y)

Here x & y are independent

Example : $p(x,y)$: $2x + 4y = 8$

u = the set of all real numbers

a) $(\forall x)(\exists y)\ p(x,y) \equiv$ true

since $2(2) + 4(1) = 4 + 4 = 8$

b) $(\exists x)(\forall y)\ p(x,y) \equiv$ false

$2x_0 + 4y = 8 \quad \forall y \in (R)$

let $x_0 = 2 \Rightarrow 2x_0 + 4y = 4 + 4y = 8$

$\Rightarrow$ false for some y.

c) $(\forall x)(\exists y)\ p(x,y) \equiv$ true

since $2x + 4y = 8 \Rightarrow 4y = 8 - 2x$

let $a = x \Rightarrow 4y = 8 - 2a$

$\Rightarrow y = \frac{8 - 2a}{4} = 2 - \frac{1}{2}a$.

Arguments and Validity.

Defn: An argument (logical deduction) is an assertion that a given set of statements $P_1, P_2, \ldots, P_n$ called premises, yield another statement Q, called Conclusion.

$P_1, P_2, P_3, \ldots, P_n \vdash Q$. (symbol)

Valid if

$(P_1 \wedge P_2 \wedge \ldots \wedge P_n) \rightarrow Q$ is a tautology

Example

1) If I sleep a lot, then I don't do any home work. If don't do any home work, then I will do well in the class.

Therefore If I sleep a lot or don't do any home work, I will do well in the class

p: I sleep a lot q: I don't do any home work

r: I will do well in the class

$p \rightarrow \neg q, \ \neg q \rightarrow r \vdash (p \vee q) \rightarrow r$

$x = z^2$

d	p	r	f → d	r → f	r ∈ (f ∩ d)	(r ∈ (f ∩ d)) → (r ∈ f) ∨ (r ∈ d)
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	T	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	F	T	T	T
F	F	F	T	T	T	T

Valid Argument

Warning In logic the ward true and valid have very different meaning : truth is taking about the statement making up an argument and validity taking about wether the Conclusion follows from the premises.

Valid Argument may have False Conclusion depending upon the truth value of the premises. likewise a invalid arguments may have a true Conclusion depending upon the truth value of the premises.

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The term set refers to well defined collection of objects that share certain property(ies).

Chapter-2: Set and Counting Theory

2.1 Basic definition and operation on set

Basic definition of set & description of set:

A) Verbal method: set of counting number less than ten, set of all countries in Africa.

B Roster / complete listing method.

a, s the set of positive factors of 36

$= \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

C) Partial listing method

es these set of all Counting number $= \{1, 2, 3, 4, \ldots\}$

the set of positive integer $= \{0, 1, 2, 3, 4, 5, \ldots\}$

D set builder method

$A = \{x \mid p(x)\}$ or $A = [x : p(x)]$

$p(x)$ - property of X

Example: 1) $\{n \mid 2n - 30$ is negative $\} \subseteq A$

2) $D = \{y \mid y^2 - y - 6 = 0\}$

Some important term in set theory

1) empty set $= \phi = \{\}$

2) Subset: $A \subseteq B$ $(\forall x)(x \in B \rightarrow x \in A)$

3) Equality set: $(\forall x)(x \in B \leftrightarrow x \in A)$ then $A = B$

4) proper subset $A \subset B$

$(\forall x)(x \in A \rightarrow x \in B) \wedge A \neq B)$

5) Power set of A: $P(A) = \{B : B \subseteq A\}$

$P(A)$ is the set whose elements are all subsets of A.

Operation on set

1) Intersection: $A \cap B = \{x : x \in A \wedge x \in B\}$

2) Union: $A \cup B = \{x : x \in A$ or $x \in B\}$

3 Relative Complement: $A - B$ or $B - A$

$A - B$ read as relative Complement of B w.r.t A.

$A - B = \{x : x \in A \wedge x \notin B\}$

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4) Absolute Complement ($A' = U - A$)

$A' = \{x : x \in U \wedge x \notin A\}$

5) Symmetric difference $A \Delta B = (A - B) \cup (B - A)$

Theorem 1: for any two set $A \cup B$

1) $(A')' = A$ 2) $A' = U - A$ 3) $A - B = A \cap B'$

4) $(A \cup B)' = A' \cap B'$

5) $(A \cap B)' = A' \cup B'$

6) $A \subseteq B \Leftrightarrow B' \subseteq A'$

Theorem 2 for any three sets A, B and C

a) $A \cup B = B \cup A$

b) $A \cap B = B \cap A$

c) $(A \cup B) \cup C = A \cup (B \cup C)$

d) $(A \cap B) \cap C = A \cap (B \cap C)$

e) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

f) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

2.2 a) Product of set (Cartesian product)

Defn: Suppose A and B are set. The Cartesian prduct of A and B denoted by $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$

b) Relation: If A and B are set; any subset of $A \times B$ is called a relation from A to B

Suppose R is relation from set A to B then $R \subseteq A \times B$.

Example: Let R be relation defined on M by $R = \{(a, b) : a, b \in M, a + 2b = 30\}$

Find R, Dom(A), Range(R), R^{-1}

Sol: $b = 1 \Rightarrow a + 2(1) = 30 \Rightarrow a = 28$

$b = 2 \quad a = 30 - 2b = 26$ so on.

In the Class $b = 3 \quad b = 30 - 6 = 24$

Function : A function is a relation in which each element of the domain Corresponding to exactly one element of the range.

Example: find domain & range of the $f(x) = (x^2 - 6x + 8)^{1/2}$

example: let $A = \{1, 2, 3\}$, $B = \{3, 5, 6, 8\}$

which of the following are functions from A to B

a) $f = \{(1,3), (2,3), (3,3)\}$ — not (B)

b) $f = \{(1,3), (2,5), (1,6)\}$ — not

c) $f = \{(1,8), (2,5)\}$ — (c)

d) $f = \{(1,6), (2,5), (3,3)\}$ — yes.

Ch-2 : set, relation & function are revision.

2.3 The basic of Counting

Counting Applications

Counting has many applications in Computer science and mathematics.

For example:

- Counting the number of operation used by an algorithm to study its time complexity
- Counting the successful outcomes of experments
- Counting all the possible outcomes of experments.

Basic Counting principles:

Two basic Counting principles

A 1). Product Rule

The product rules applies when a procedures are makeup of seperate tasks.

B 2) Sum Rule: Assume a tasks can be done either in one n_1 ways or in one of n_2 ways

Assume none of the set of n_1 ways is the same as any of the set n_2 ways so there are $n_1 + n_2$ ways to do the task.

Detail of product Rule.

Suppose that a procedures can be broken into a sequence of two tasks

Assume there are n_1 ways to do the first task

Assume there are n_2 ways to do the second task

So there are $n_1 \cdot n_2$ ways to do the procedures.

Example 1
A new company with just two employees rents a floor of a building with 12 offices. How many ways are there to assign the offices to these two employees?

Soln Break procedure into tasks
Task1: Assign an office to employee 1
Task2: Assign an office to employee 2
Count diff ways of doing each task
T_1 in 12 diff ways
T_2 in 11 diff ways
Product Rule: There are $12 * 11 = 132$ ways to assign offices to two employees.

Example 2:
The Format of telephone numbers in Ethiopia is specified by numbering plan
A) Let X denote a digit between 0 and 9
B) Let N denote a digit bln 2 and 9
C) Let Y denote a digit bln 0 and 1
In old plan, The format of telephone number is NYX-NNY-XXXX
In new plan, the format of telephone number is NXX-NXX-XXXX
How many telephone numbers are possible under the old and new plan

Soln
X has 10 diff possible
N has 8 diff possible
Y has 2 diff possible
oldplan = $(8 \cdot 2 \cdot 10) \cdot (8^2 \cdot 10) \cdot (1 \cdot 10^4)$
$= 1,024,000,000$
Newplan: $(8 \cdot 10^2)(8 \cdot 10^2) \cdot (10^4)$
$= 6,400,000,000$

The Sum Rule

Assume a task can be done in one of n_1 ways or in one of n_2 ways

Assume none of the set of n_1 ways is the same as any of the set n_2 ways

So, there are $n_1 + n_2$ ways of doing the task

Example. A student can be choose a Computer project from one three lists. The three lists contains 23, 15 and 19 possible projects. No project is on more than one list.

How many possible projects are there to choose from?

Solu: The student can choose a project from the first, second or the third list.

By product rule = $23 + 15 + 19 = 57$ ways to choose the projects.

The Product Rule and The Sum Rule

Some complicated Counting problems can be solved using both the product rule and the sum Rule.

Example: In Computer system, each user has a password

- the password must be six to eight character long
- ✓ Each character is an upper case or digit
- ✓ Each password must contain at least one digit

Solution

Task 1: Choose six character long password

Task 2: Choose seven character long password

Task 3: Choose a eight character password
with the number of task 1 : the number of pass
words of six characters where each character
can be an Upper Case letter or a digit is 36^6
excluding the number of six character long
password with no digit which is 26^6
Task 1 can be done in $= 36^6 - 26^6$
$= 1,867,866,560$ ways.

Similar ways
$\text{Task}_2 = 36^7 - 26^7 = 70,332,353,920$ ways
$\text{Task}_3 = 36^8 - 26^8 = 2,612,282,842,880$
ways.

By sum Rule = $\text{Task}_1 + \text{Task}_2 + \text{Task}_3$
$= 2,684,483,063,360$ ways.

2.4 Permutation and Combination.

Permutations

Definition (permutation of set.)

Given set S, a permutation of S, is arrangement of the element of S in specific order without repetition.

Note: Recall that the set S itself cannot have repeated elements.

Theorem: (The number of permutation of set S)

The number of permutation of set S of size $n(S) = n$ denoted by nPn is

$$nPn = n(n-1)(n-2)\cdots 2.1 = n!$$

Definition (r-permutation of set S)

Given a set S of size $n(S) = n$, an r-permutation ($r \leq n$) of S is an arrangement of r element of S in specific order without repetition.

Theorem: (Number of permutation of set S)

The number of r-permutation of set S of size $n(S) = n$ denoted by nPr is

$$nPr = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1)$$

Note $nPn = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$

Example

Example

Theorem 3: The number of permutations of n different objects taken r at a time where repetition is allowed is n^r

Example 1: Find number of 4 letter words with or without meaning, which can be formed from the words ROSE, where repetition of letters not allowed

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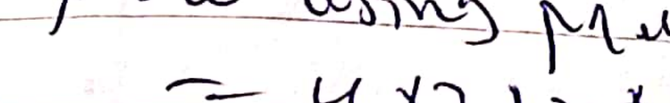
Solution: Filling in 4 vacant places



by four letters

The 1st vacant (Places) can be filled in 4 dlt ways by any one of R, O, S, E

Second place can be filled by any one of the remains 3 letters in 3 dlt ways

- third place can be filled in 2 remaining letter in 2 dlt ways

- fourth place can be filled in remains one letter in one ways)

Now using Multiplication priciple we have

$= 4 \times 3 \times 2 \times 1 = 24$ different ways)

$= nPn = n! = 4! = 4 \times 3 \times 2 \times 1 = 24$

If repetition allowed: each vacant places can be filled in successive in 4 dlt ways)

Hence required number of words $= 4 \times 4 \times 4 \times 4$

$= 4^4 = 256$ words) can be formed.

If we have to determine the number of 3 letters words with or without meaning from word ROSE without repetition

$n = 4, r = 3 \rightarrow {}^4P_3$

$= \frac{4!}{(4-3)!} = \frac{4!}{1!} = 4! = 4 \times 3 \times 2 \times 1 = 24$

Fillins three vacant

4 3 2

$= 4 \times 3 \times 2 = 24$ dlt words)

If repetition allowed: for each places we have 4 chances $= 4 \times 4 \times 4 = 4^3 =$

$= 16 \times 4 = 64$ dlt ways)

Exercise: try to list all words.

Permutation when all the objects are not distinct objects.

1) Suppose we have to find number of ways of rearranging the letters of word ROOT. There are 2 Os which are same find. Therefore the required number of permutation $= \frac{4P4}{2!}$

$= \frac{4!}{(4-4)!\,2!} = \frac{4!}{2!} = 4 \times 3 = 12$

2) Example: find number of ways of rearranging the letters of the word INSTITUTE

Sol: In this case there are n = 9 letters in which I appears 2 times and T appears 3 times

total number of permutation

$= \frac{9!}{2!3!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{2 \times 3 \times 2}$

$= \frac{9 \times 7 \times 8 \times 6 \times 5 \times 4}{2} = \frac{30240}{\text{diff ways}}$.

Combination

Definition (r-Combination of a set)
Given set S of size $n(S)=n$, an r-combination ($r \leq n$) of S is a subset of k distinct elements of S (no repetition of element to allowed.)

Note: The arrangement (order) of the elements in the subset does not matter

Theorem 4: (Number of r-combination)
The number of r-Combination of set of size $n(S)=n$ denoted by nCr is

$$nCr = \binom{n}{r} = \frac{nPr}{r!} = \frac{n!}{r!(n-r)!}$$

Example: A Committe of 3 persons is to be Constituted from a group of 2 men and 3 women. In how many ways this can be done? How many of these committe would consist of 1 man and 2 women?

Soln I) 5 diff person = 2+3 = 5
taken 3 at a time the required number of ways $= 5C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!}$

$$= \frac{4 \times 5}{2} = \frac{20}{2} = 10$$

I) one man can be selected from 2 men in $2C_1 = \frac{2!}{1!(2-1)!} = \frac{2}{1} = 2$

and 2 women can be selected from 3 women in $3C_2 = \frac{3!}{2!(3-2)!}$

$$= \frac{3!}{2!1!} = \frac{3 \times 2}{2} = 3$$

total $= 3C_2 \times 2C_1 = 3 \times 2 = 6$ all ways

Example 2 : A Group (Consists) of 4 girls and 7 boys
In how many ways can a team 5 member be selected if the team has
i) no girl ?
ii) atleast one boy and one girls ?
iii) atleast three girls ?

Soln i) Since the team will not include any girl therefore the only boy to be selected in $^7C_5 = \frac{7!}{5!(7-5)!} = \frac{7!}{5!2!}$

$= \frac{7!}{5!2!} = \frac{7\times6\times5!}{5!\times2!} = \frac{7\times6}{2} = 21$ different ways

ii) Since atleast one boy & 1 girl are to be there in the team, therefore the team can be consists of
a) 1 boy and 4 girl $= {}^7C_1 \times {}^4C_4$
b) 2 boy and 3 girls $= {}^7C_2 \times {}^4C_3$
c) 3 boy and 2 girls $= {}^7C_3 \times {}^4C_2$
d) 4 boy and 1 girls $= {}^7C_4 \times {}^4C_1$

therefore the required number of ways
$= {}^7C_1 \times {}^4C_4 + {}^7C_2 \times {}^4C_3 + {}^7C_3 \times {}^4C_2 + {}^7C_4 \times {}^4C_1$
$= 7 + 84 + 210 + 140 = 441$

(iii) Since the team consist of atleast a three (3) girls. therefore the team consist of
a) 3 girls and 2 boys $= {}^4C_3 \times {}^7C_2$
b) 4 girls and 1 boys $= {}^4C_4 \times {}^7C_1$

Therefore the required number of ways
$= {}^4C_3 \times {}^7C_2 + {}^4C_4 \times {}^7C_1$
$= 84 + 7 = 91$ ways.

Summary

ORder matters		
yes	With out rep	With repetition
	Permutation	
	$P(n,r) = \frac{n!}{(n-r)!}$	$P^*(n,r) = n^r$
no		With repetition
$n C r = \binom{n}{r}$	Combination	
	$C(n,r) = \frac{n!}{r!(n-r)!}$	$C^*(n,r) = \frac{(n+r-1)!}{r!(n-1)!}$

- Permutation, Combination and probablity.
- Example of Combination with repetition

Combination with repetition

Given n objects we want to select r objects with replacement

$$= \binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

Example: If there are 5 flavor of icecream and you have (can) 3 Scoops of icecrem, How many Combinations Can you have?

repeat flavors

Soln $n = 5$, $r = 3$

$$\frac{(5+3-1)!}{3!(5-1)!} = \frac{7!}{3!4!} = 35 \text{ Combination}$$

Fundamental Theorem

probality example

$$P = \frac{\#\ \text{successes}}{\text{total Outcomes}}$$

Example: you and 3 Freinds are Running for a 4 persons Committe, 100 person are running. What is the probablity that you and your 3 friends won the 4 seats?

Solution: $\frac{4C4}{100C4} = \frac{1}{3,921,225}$

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Fundamental Theorem
probality example

$$P = \frac{\#\ \text{successes}}{\text{total Outcomes}}$$

Example: you and 3 Freinds are Running for a 4 person Committe, 100 person are running. What is the probablity that you and your 3 friends win the 4 seats?

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Binomial theorem

Binomial theorem for positive integral Indices

$(a+b)^0 = 1$

$(a+b)^1 = a+b$, $a+b \neq 0$

$(a+b)^2 = a^2 + 2ab + b^2$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

We observe that:

1) total number of terms in the expansion one more than the noted

2) powers of the first quantity goon decreasing by 1 where as the powers of the second quantity increases by 1 In the successive term

3) In each term of the expansion the sum of the indices of a and b is the same as equal to the indices of a+b

4) The coefficient of each term is obtained by pascal triangle

Index	Coefficient
0	1
1	1 1
2	1 2 1
3	1 3 3 1
4	1 4 6 4 1
5	1 5 10 10 5 1
6	1 6 15 20 6 1
7	1 21 35 26 7 1
8	1 28 56 70 33 8 1
9	1 23 79 118 84 45 9 1
10	1 24 102 197 252 135 30 1

And soon.

Example: Let us expand $(2x+3y)^7$ by using Pascal Rule

Index 7 is 1 7 10 10 7 1

$= (2x)^7 + 7(2x)^6(3y) + 10(2x)^3(3y)^2 + 10(2x)^2(3y)^3 + 7(2x)(3y)^6 + (3y)^7$

$= 3rx^7 + 240x^4y + 720x^3y^2 + 1080x^2y^3 + 810xy^4 + 243y^7$

Note The number in the pascal triangle is equal to nCr

$nCr = \frac{n!}{r!(n-r)!} = \binom{n}{r}$

$0 \le r \le n$ n is non negative integer

Binomial theorem

$(a+b)^n = \sum_{k=0}^{n} nC_k\, a^{n-k} b^k$

Example: expand $(x^2 + \frac{3}{x})^4$, $x \neq 0$

Solution using binomial theorem

$(x^2 + \frac{3}{x})^4 = 4C_0(x^2)^4 + 4C_1(x^2)^3(\frac{3}{x}) + 4C_2(x^2)^2(\frac{3}{x})^2 + 4C_3(x^2)(\frac{3}{x})^3 + (4C_4)(x^2)^0(\frac{3}{x})^4$

$= \text{Note } 4C_0 = 1, \quad 4C_4 = 1$

$\Rightarrow (x^2)^4 + 4x^6 \cdot \frac{3}{x} + 6x^4(\frac{9}{x^2}) + 4x^2 \cdot \frac{27}{x^3} + \frac{81}{x^4}$

$= x^8 + 12x^5 + 54x^2 + \frac{108}{x} + \frac{81}{x^4}$.

General term and middle term

A) In binomial theorem expansion for $(a+b)^n$ we observe that the first term ⇒

$nC_0 a^n$ — 1st term

$nC_1 a^{n-1}b$ — 2nd term

$nC_r a^{n-2}b^r$ — 3rd term

$nC_3 a^{n-3}b^3$ — 4th term and so on

Generally the $(r+1)$ term ⇒

$nC_r a^{n-r}b^r$

B) $T_{r+1} = nC_r a^{n-r}b^r$

- Middle term in expansion $(a+b)^n$

We have

i) If n-even the total number of expansion will be $n+1 =$ odd

Hence the middle term ⇒ $\left(\frac{n+2}{2}\right)^{th}$ th

$= \left(\frac{n}{2}+1\right)^{th}$ term

For example in expansion of $(x+y)^8$

the middle term is $\left(\frac{8}{2}+1\right) = 5^{th}$ term

the Coefficients $T_5 = T_{r+1} =$

$r+1 = 5 \Rightarrow r = 4$

$5C_4 (x)^4 (2y)^4$

$= 2^4 \cdot 5C_4 x^4 y^4$

$= 16 \times 5C_4$

$5C_4 = \frac{5!}{4!(1)!} = \frac{5!}{4!} = 5$

$= 5 \times 16 = 80$

$\Rightarrow 80 x^4 y^4$

II) If n = odd then $n+1$ = even so there will be two middle term $\left(\frac{n+1}{2}\right)$ and $\left(\frac{n+1}{2}+1\right)^{th}$ term

Example: In the expansion $(2x-7)^7$ the middle term are $\left(\frac{7+1}{2}\right)^{th}$ term & $\left(\frac{7+1}{2}+1\right)^{th}$ term

$=$ 4th term and 5th term.

Example: Find a if the 17th and 18th term are equal in expansion $(x+a)^{20}$

$T_{17} = T_{r+1} \Rightarrow r = 16$

$T_{18} = T_{r+1} \Rightarrow r = 17$

$T_{17} = T_{16+1} = {}^{20}C_{16}(x)^{20-16} a^{16}$

$T_{18} = T_{17+1} = {}^{20}C_{17}(x)^{20-17} a^{17}$

which are equal to

$T_{17} = {}^{20}C_{16} x^{4} \cdot a^{16}$

$T_{18} = {}^{20}C_{17} \cdot x^{3} a^{17}$

$T_{17} = T_{18} \Rightarrow {}^{20}C_{16} \cdot x^{4} \cdot a^{16} = {}^{20}C_{17} \cdot x^{3} a^{17}$

$\Rightarrow \frac{{}^{20}C_{16} \cdot x^{4}}{{}^{20}C_{17} \cdot x^{3}} = \frac{a^{17}}{a^{16}}$

$a = \frac{{}^{20}C_{16} \cdot x^{4}}{{}^{20}C_{17} \cdot x^{3}} = \frac{1}{}$

Exercise ——— Exercise

1) The sum of the coefficient of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0$, n is natural numbers > 11 ?
find the term of the expansion Contains x^3.

2) Find the term independent of x in the expansion of $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^6$

) find the 13th term in the expansion of $\left(9x - \frac{1}{3\sqrt{x}}\right)^{18}$, $x \neq 0$.

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Ch-3

Advanced Counting techniques

3.1 Recurrence Relation (RR)

A) Linear Homogeneous Recurrence Relation

B) Solving linear homogeneous Recurrence Relation with Constant Coefficient

C) Solving linear Recurrence relation with Constant Coefficients.

A) Linear Homogeneous Recurrence Relation

Definition: Linear homogeneous recurrence relation of degree k with Constant coefficient is a recurrence Relation of the form

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k \cdot a_{n-k}$

where $C_1, C_2, \ldots, C_k$ are real numbers with $C_k \neq 0$.

B) Solving Linear Homogeneous Recurrence relations.

The basic approach is to look for the solution of the form

$a_n = r^n$, where r is Constant

Note that $a_n = r^n$ is solution to recurrence relation $a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ if and only if

$r^n = C_1 r^{n-1} + C_2 r^{n-2} + \cdots + C_k r^{n-k}$

$\rightarrow r^k = C_1 r^{k-1} + C_2 r^{k-2} + \cdots + C_k$

$\rightarrow r^k - (C_1 r^{k-1} + C_2 r^{k-2} + \cdots + C_k) = 0$

is called Characteristic equation of the Recurrence Relation.

Theorem: Let C_1 and C_2 be real numbers. Suppose $r^2 - C_1 r - C_2 = 0$ has two distinct roots r_1 and r_2. Then the sequence $\{a_n\}$ is a solution of the recurrence relation (RR)

$a_n = C_1 a_{n-1} + C_2 a_{n-2}$ if and only if

$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$

where α_1, α_2 are constants

Example: (using theorem). What is the solution of Recurrence Relation

$a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2, a_1 = 7$?

Solution: Char. eqn. is $r^2 - r - 2 = 0$

Its roots are $r_1 = 2, r_2 = -1$

$a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, for some α_1 & α_2

To find constants α_1 & α_2

$a_0 = \alpha_1 2^0 + \alpha_2 (-1)^0 = \alpha_1 + \alpha_2 = 2$

$a_1 = \alpha_1 (2)^1 + \alpha_2 (-1)^1 = 7$

$\Rightarrow 2\alpha_1 + -\alpha_2 = -7$

$\rightarrow \alpha_1 + \alpha_2 = 2$ } $\Rightarrow \alpha_1 = 3$

$2\alpha_1 + (-\alpha_2) = 7$ } $\alpha_2 = -1$

Hence the solution is the sequence

$\{a_n\}$ with $a_n = 3.(2)^n - (-1)^n$

$a_n = 3.2^n + (-1)^{n+1}$

Example 2: Solve RR $f_n = f_{n-1} + f_{n-2}$

$f_0 = 0, f_1 = 1$

Solution: Char. equation $r^2 - r - 1 = 0$

$\Rightarrow r_1 = \frac{1+\sqrt{5}}{2}, r_2 = \frac{1-\sqrt{5}}{2}$

By thm 1: $f_n = \beta_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \beta_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$

for some Constant β_1 & β_2

Using initial Condition $f_0 = 1, f_2 = 1$

$\Rightarrow \beta_1 + \beta_2 = 0 = f_0$

$\left(\frac{1+\sqrt{5}}{2}\right)\beta_1 + \left(\frac{1-\sqrt{5}}{2}\right)\beta_2 = 1 = f_1$

$\Rightarrow \beta_1 = \frac{1}{\sqrt{5}}$ & $\beta_2 = \frac{-1}{\sqrt{5}}$

Hence $f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$

Theorem (Solution if repeated root)
Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has one repeated root r_0. Then the sequence $\{a_n\}$ is a solution to the RR $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \beta_1 \cdot r_0^n + \beta_2 n r_0^n$. for $n = 0, 1, 2, \ldots$ where β_1 and β_2 are constants.

Example: What is the solution of the RR $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1, a_1 = 6$

Solution: Charac. equation: $r^2 - 6r + 9 = 0$
the only root $r = 3 = 3$
Therefore the solution of the RR is

$a_n = \beta_1 (3)^n + \beta_2 n (3)^n$ where
β_1, β_2 are constants

$a_0 = \beta_1 (3)^0 + \beta_2 (0) \cdot 3^n = \beta_1 = 1$
$a_1 = \beta_1 (3) + \beta_2 3 = 6$
$\rightarrow 3 + 3\beta_2 = 6 \rightarrow \beta_2 = 1.$
Hence
$a_n = 3^n + n3^n$

Theorem 3: Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation
$r^k - (c_1 r^{k-1} + \ldots + c_k) = 0$ has k distinct roots $r_1, r_2, \ldots, r_k$.
Then a sequence $\{a_n\}$ is a solution of the RR

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ d.f

$a_n = \beta_1 r_1^n + \beta_2 r_2^n + \cdots + \beta_k r^n$ for
$n = 0, 1, 2, \ldots$ where
$\beta_1, \beta_2, \ldots \beta_k$ are constants

Theorem 4: Let $C_1, C_2, \ldots, C_k$ be real numbers. Suppose that the characteristic equation

$r^k - C_1 r^{k-1} - C_2 r^{k-2} - \cdots - C_k = 0$

has t distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities
$m_1, m_2, \ldots, m_t$ respectively
so that $m_i \geq 1$ for $i = 1, 2, \ldots, t$
and $m_1 + m_2 + \cdots + m_t = k$
k is degree of the RR.
Then a sequence $\{a_n\}$ is a solution of the RR

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$

if and only if

~~$a_n = ($~~ $\beta_{1,0}$

$a_n = (\beta_{1,0} + \beta_{1,1} n + \cdots + \beta_{1,m_1-1} n^{m_1-1}) r_1^n$
$+ (\beta_{2,0} + \beta_{2,1} n + \cdots + \beta_{2,m_2-1} n^{m_2-1}) r_2^n$

$+ \cdots + \cdots$

$+ (\beta_{t,0} + \beta_{t,1} n + \cdots + \beta_{t,m_t-1} n^{m_t-1}) r_t^n$

for $n = 0, 1, 2, \ldots$ and $\beta_{i,j}$ are
constants for $1 \leq i \leq t$ and

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ d.f

$Q_n = \beta_1 r_1^n + \beta_2 r_2^n + \cdots + \beta_k r^n$ for
$n = 0, 1, 2, \ldots$ where
$\beta_1, \beta_2, \ldots \beta_k$ are constants

Theorem 4: Let $C_1, C_2, \ldots, C_k$ be real numbers. Suppose that the characteristic equation

$r^k - C_1 r^{k-1} - C_2 r^{k-2} - \cdots - C_k = 0$

has t distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities
$m_1, m_2, \ldots, m_t$ respectively
so that $m_i \geq 1$ for $i = 1, 2, \ldots, t$
and $m_1 + m_2 + \cdots + m_t = k$
k is degree of the RR.
Then a sequence $\{a_n\}$ is a solution of the RR

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$

if and only if

~~$a_n = (\beta_{1,0}$~~

$a_n = (\beta_{1,0} + \beta_{1,1} n + \cdots + \beta_{1,m_1-1} n^{m_1-1}) r_1^n$
$+ (\beta_{2,0} + \beta_{2,1} n + \cdots + \beta_{2,m_2-1} n^{m_2-1}) r_2^n$
$+ \cdots + \cdots + \cdots +$
$+ (\beta_{t,0} + \beta_{t,1} n + \cdots + \beta_{t,m_t-1} n^{m_t-1}) r_t^n$

for $n = 0, 1, 2, \ldots$ and β_{ij} are
constants for $1 \leq i \leq t$ and
$0 \leq j \leq m_i - 1$.

Example (Theme)

Solve the RR $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial Condition $a_0 = 1, a_1 = -2$ and $a_2 = -1$.

Solution

$$r^3 + 3r^2 + 3r + 1 = 0$$

$r = -1$ is the only solution

$$\Rightarrow a_n = \beta_1 r^n + n\beta_2 r^n + n^2 \beta_3 r^n$$

$a_0 =$ $\beta_1 = 1$

$a_1 =$ $\beta_1(-1) + (1)(\beta_2)(-1) + (1)\beta_3(-1) = -2$

$-1 - \beta_2 - \beta_3 = -2$

$\beta_2 + \beta_3 = 1 \quad -(1)$

$a_2 = -1 =$ $\beta_1 + 2\beta_2 + 4\beta_3$

$-1 = 1 + 2\beta_2 + 4\beta_3$

$-2 = 2\beta_2 + 4\beta_3 \quad -(2)$

From (1) & (2)

$\beta_2 + \beta_3 = 1$

$2\beta_2 + 4\beta_3 = -2$

$\Rightarrow \beta_2 = 3$

$\beta_3 = -2$

Hence $a_n = (1 + 3n - 2n^2)(-1)^n$

Hence $a_n = 1 + 3n - 2n^2$ if n = even

$a_n = -(1 + 3n - 2n^2)$ if n = odd.

Exercise:

2)

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linear non homogeneous RR with Constant Coefficients.

Defn: A linear non H RR with Constant Coefficients is a RR of the form

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k} + F(n)$ where $C_1, C_2, \cdots C_k$ are ~~Real~~ Real numbers and $F(n)$ is a function not identically zero. depending only on n

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ is called the associated homogeneous RR.

A particular solution of RR is a sequence that satisfies the ~~recurrence~~ recurrence equation

However it may or may not satisfy the initial conditions.

Example: Solve the recurrence Relation $a_n = 3a_{n-1} + 2n$, with initial condition $a_1 = 3$

Solution: H RR: $a_n = 3a_{n-1}$

$r^n = 3r^{n-1}$

$r^n - 3r^{n-1} = 0$

$r - 3 = 0$

$r = 3$

$a_n = \beta r^n$

$a_1 = \beta r = 3$

(1) Solving Recurrence Relation by generating fn (1)

Generating function

$$G(x) = g_0 + g_1 x + g_2 x^2 + g_3 x^3 + \cdots = \sum_{k=0}^{\infty} g_k x^k$$

Sl. No:	General term of sequence g_k	Generating function $G(x)$
1	1	$\frac{1}{1-x}$
2	$k+1$	$\frac{1}{(1-x)^2}$
3	k	$\frac{x}{(1-x)^2}$
4	$k(k+1)$	$\frac{2x}{(1-x)^3}$
5	$(k+1)(k+2)$	$\frac{2}{(1-x)^3}$
6	a^k	$\frac{1}{1-ax}$

Example: Solve the RR by generating function

$a_{n+2} - 3a_{n+1} + 2a_n = 0$, $a_0 = 2$, $a_1 = 3$

multiply the equation with x^n and summ from 0 to ∞.

Expanding we get

$$\sum_{n=0}^{\infty} a_{n+2} x^n - 3 \sum_{n=0}^{\infty} a_{n+1} x^n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

Expanding ... we get

$$(a_2 + a_3x + a_4x^2 + \dots) - 3(a_1 + a_2x + a_3x^2 + \dots) + 2(a_0 + a_1x + a_2x^2 + \dots) = 0$$

We know $G(x) = a_0 + a_1x + a_2x^2 + \dots$

we manipulate above equation to obtain expression of in terms of G

For example $a_2 + a_3x + a_4x^2 + \dots$

$$= \frac{a_2x^2 + a_3x^3 + a_4x^4 + \dots}{x^2}$$

$$= \frac{a_0 + a_1x + a_2x^2 + a_3x^3 + \dots - a_0 - a_1x}{x^2}$$

$$= \frac{G(x) - a_0 - a_1x}{x^2}$$

Similarly we get

$$\frac{G(x) - a_0 - a_1x}{x^2} - 3\left(\frac{G(x) - a_0}{x}\right) + 2G(x) = 0$$

Substituting the intial values

$a_0 = 2$, $a_1 = 3$

$$\frac{G(x)-2-3x}{x^2} - 3\frac{G(x)-2}{x} + 2G(x) = 0$$

$$G(x) = \frac{2-3x}{2x^2-3x+1}$$

using partial fraction decomposition

$$\frac{2-3x}{1-3x+2x^2} = \frac{1}{1-x} + \frac{1}{1-2x}$$

using the table / we get

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 1^n x^n + \sum_{n=0}^{\infty} 2^n x^n$$

Reducing the expression / we get

$$a_n = 1 + 2^n$$

Example: Solve RR $a_n = a_{n-1} + 2a_{n-2}$

Example 2 $a_0 = 3$ and $a_2 = 7$

multiply the equation with x^n and sum from $n = 0$ to ∞

$$\sum_{n=2}^{\infty} a_n x^n = \sum_{n=2}^{\infty} a_{n-1} x^n + 2\sum_{n=2}^{\infty} a_{n-2} x^n$$

keeping the exponent of x equal to the suffix of a inside the summation sign.

$$\sum_{n=2} a_n x^n = x \sum_{n=2} a_{n-1} x^{n-1} + 2x^2 \sum_{n=2} a_{n-2} x^{n-2}$$

Expanding -, we set

$$(a_2 x^2 + a_3 x^3 + \cdots) = x(a_1 x + a_2 x^2 + \cdots) + 2x^2(a_0 + a_1 x + a_2 x^2 + \cdots)$$

$$\Rightarrow a_2 x^2 + a_3 x^3 + \cdots - a_0 - a_1 x + a_0 + a_1 x$$

$$= a_2 x^2 + a_3 x^3 + \cdots \quad = G(x) - a_0 - a_1 x$$

$$(a_0 + a_1 x + a_2 x^2 + \cdots - a_0) x. = (G(x) - a_0) x$$

we set interms of Generating function

$$\rightarrow G(x) - a_0 - a_1 x = x(G(x) - a_0) + 2x^2(G(x))$$

$$G(x) - a_0 - a_1 x = x G(x) - a_0 x + 2x^2 G(x)$$

$$G(x) - x G(x) - 2x^2 G(x) = a_0 + a_1 x$$

$$G(x)(1 - x - 2x^2) = 3 + 4x$$

$$G(x) = \frac{3 + 4x}{1 - x - 2x^2}$$

using partial fraction decomposition

$$\frac{3 + 4x}{1 - x - 2x^2} = \frac{10}{3(1 - 2x)} - \frac{1}{3(1 + x)}$$

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$$G(x) = \frac{10}{3(1-2x)} - \frac{1}{3(1-x)}$$

using table result

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$$G(x) = \frac{10}{3}\sum_{n=0}^{\infty} 2^n x^n - \frac{1}{3}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$\sum_{n=0}^{\infty} a_n x^n = \frac{10}{3}\sum_{n=0}^{\infty} 2^n x^n - \frac{1}{3}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$\rightarrow a_n = \frac{10}{3}(2)^n + \frac{-1}{3}(-1)^n$$

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Ch - 3: Advanced Counting Techniques

Solving RR using Generating function

Examples: Solve RR : $a_n - 3a_{n-1} = n$, $n \geq 1$

$a_0 = 1$

solution: $a_n = a_n^h + a_n^p$

Let us write few first equation

$n \geq 1$ $\quad a_1 - 3a_0 = 1$

$n = 2$ $\quad a_2 - 3a_1 = 2$

$n = 3$ $\quad a_3 - 3a_2 = 3$

Multiply the nth equation by x^n

$n = 1$ $\quad a_1 x - 3a_0 x = 1x$

$n = 2$ $\quad a_2 x^2 - 3a_1 x^2 = 2x^2$

$n = 3$ $\quad a_3 x^3 - 3a_2 x^3 = 3x^3$

Let us add these all up

$$\sum_{n \geq 1} a_n x^n = a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

$$+ 3a_0 x + 3a_1 x^2 + 3a_2 x^3 + \cdots$$

$$= \sum_{i=0} 3a_i x^{i+1} = 3x \sum_{i=0} a_i x^i$$

$$= 3 \sum_{i=1} a_{i-1} x^i = 3x \sum_{n \geq 0} a_n x^n$$

$$1x + 2x^2 + 3x^3 + \cdots = \sum_{n=1} n x^n$$

set $f(x) = \sum_{n \geq 0} a_n x^n$, then do the same for $f(x)$

$$\sum_{n \geq 1} a_n x^n - 3 \sum_{n \geq 1} a_{n-1} x^n = \sum_{n \geq 1} n x^n$$

$$(f(x) - a_0) - 3x f(x) = \sum_{n \geq 1} n x^n$$

$$\sum_{n=1} nx^n = x + 2x^2 + 3x^3 + 4x^4 + \cdots$$

$$= x(1 + 2x + 3x^2 + \cdots$$

$$\sum_{n\geq 1} nx^n = x\frac{d}{dx}\left(1 + x + x^2 + x^3 + \cdots\right)$$

$$= x\frac{d}{dx}\left(1 + x + x^2 + x^3 + x^4\right)$$

$$= x\frac{d}{dx}\left(\frac{1}{1-x}\right)$$

$$= x\left(\frac{1}{(1-x)^2}\right) = \frac{x}{(1-x)^2}$$

Hence $f(x) - a_0 = 3xf(x) + \dfrac{x}{(1-x)^2}$

$$f(x) - 3xf(x) = \frac{x}{(1-x)^2} + a_0$$

$$(1-3x)\,f(x) = \frac{x}{(1-x)^2} + 1 = \frac{x + (1-x)^2}{(1-x)^2}$$

$$f(x) = \frac{x + (1-x)^2}{(1-3x)(1-x)^2}$$

$$= \frac{x + 1 - 2x + x^2}{(1-3x)(1-x)^2}$$

$$f(x) = \frac{x^2 - x + 1}{(1-x)^2(1-3x)}$$

next step is partial fraction

$$\frac{x^2 - x + 1}{(1-x)^2(1-3x)} = \frac{A}{(1-x)} + \frac{B}{(1-x)^2} + \frac{C}{(1-3x)}$$

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$x^2 - x + 1 = A(1-x)(1-3x) + B(1-3x) + C(1-x)^2$

$= A(1 - 4x + 3x^2) + B(1-3x) + C(1 - 2x + x^2)$

$= A + B + C + (-4A - 3B - 2C)x + (3A + C)x^2$

$A + B + C = 1$

$-4A - 3B - 2C = -1$

$3A + C = 1$

} System of linear equations

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 \\ -4 & -3 & -2 \\ 3 & 0 & 1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} A \\ B \\ C \end{pmatrix}}_{x} = \underbrace{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}}_{b}$$

$Ax = b$

Augment form $[A|b]$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ -4 & -3 & -2 & -1 \\ 3 & 0 & 1 & 1 \end{pmatrix} \quad R_2 \to -1R_2$$

$R_2 \to R_2 - 4R_1$

$R_3 \to R_3 - 3R_1$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -3 \\ 0 & -3 & -2 & -2 \end{pmatrix}$$

$R_1 \to R_1 + R_2$

$R_3 \to R_3 + 3R_2$

$$\begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 7 \end{pmatrix}$$

$4C = 7 \Rightarrow C = 4/7$

$A - C = -2$

$A - 4/7 = -2$

$A = -2 + 4/7 = \frac{-14+4}{7} = -10/7$

$B + 2C = 3$

$B = 3 - 2C = 3 - \frac{8}{7} = \frac{13}{7}$

$$\frac{-10/7}{(1-x)} + \frac{+13/7}{(1-x)^2} + \frac{4/7}{(1-3x)}$$

$$= \frac{-10}{7}[x^n]\frac{1}{1-x} + \frac{13}{7}([x^n]\frac{1}{(1-x)^2} + \frac{4}{7}\frac{[x^n]}{(1-3x)}$$

$$= \frac{-10}{7}(1)^n + \frac{13}{7}(n+1) + \frac{4}{7}(3)^n$$

$$a_n = -10/7 + \frac{13}{7} + \frac{13}{7}n + \frac{4}{7}(3)^n$$

$$a_n = 3/7 + \frac{13}{7}n + \frac{4}{7}(3)^n$$

$$a_0 = 3/7 + 4/7 = 1$$

$$a_1 = 3/7 + 13/7 + \frac{4}{7} = \frac{28}{7} = 4$$

$a_2 =$

$a_n = 3a_{n-1} + n$

$a_1 = 3(1) + 1 = 4$

$a_2 = 3(4) + 2 = 12 + 2 = 14$

$a_3 = 3(14) + 3 = 42 + 3 = 45$

Examples

Final Exam

Workout Problem (Answersheet)

Generating Function solve the RR

$a_{n+1} = 3a_n - 7(n+1) + 7.2^n$ with $a_0 = 6$

Solution: $G(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots$

Use both side the Capital summation and x^n (multiply x^n)

$\sum_{n=0} a_{n+1}x^n = \sum_{n=0} 3a_n x^n - \sum_{n=0} 7(n+1)x^n + \sum_{n=0} 7.2^n x^n$

$\sum_{n=0} a_{n+1}x^n = \frac{a_1x + a_2x^2 + a_3x^3 + \cdots}{x} = a_1 + a_2x + a_3x^2 + \cdots$

$= \frac{G(x) - a_0}{x} = a_1 + a_2x + a_3x^n + \cdots \quad \cdots (1)$

$= 3G(x)$

$\sum_{n=0} 3a_n x^n = 3a_0 + 3a_1x + 3a_2x^2 + 3a_3x^3 + \cdots \quad \cdots (2)$

$\sum_{n=0} 7(n+1)x^n = \cdots 7\sum_{n=0} (n+1)x^n = \frac{7}{(1-x)^2} \quad \cdots (3)$

$\sum_{n=0} 7.2^n x^n = 7\sum_{n=0} (2x)^n = \frac{7}{1-2x} \quad \cdots (4)$

Also Collect all terms according to its position

$\frac{G(x) - a_0}{x} = 3G(x) - \frac{7}{(1-x)^2} + \frac{7}{(1-2x)}$

$G(x) - a_0 = 3xG(x) - \frac{7x}{(1-x)^2} + \frac{7x}{(1-2x)}$

$G(x) - 3xG(x) = a_0 - \frac{7x}{(1-x)^2} + \frac{7x}{(1-2x)}$

$G(x)(1-3x) = a_0 - \frac{7x}{(1-x)^2} + \frac{7x}{(1-2x)}$

$G(x) = \frac{a_0}{1-3x} - \frac{7x}{(1-x)^2(1-3x)} + \frac{7x}{(1-x)(1-3x)}$

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common factor $= (1-3x)(1-x)^2(1-2x)$

$$= \frac{a_0(1-2x)(1-x)^2 + 5x(1-2x) + 7x(1-x)^2}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0(1-2x)(1-2x+x^2) - 5x(1-2x) + 7x(1-x)^2}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0\left((1-2x+x^2) - 2x(1-2x+x^2)\right) + 5x(1-2x) + 7x(1-x)^2}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0(1-4x+5x^2-2x^3) - 5x(1-2x) + 7x(1-x)^2}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0(1-4x+5x^2-2x^3) - 5x + 10x^2 + 7x(1-2x+x^2)}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0(1-4x+5x^2-2x^3) - 5x + 7x + 10x^2 - 14x^2 + 7x^3}{(1-3x)(1-x)^2(1-2x)}$$

$$= \frac{a_0 + (2-4a_0)x + (5a_0-4)x^2 + (7-2a_0)x^3}{(1-3x)(1-x)^2(1-2x)}$$

use partial fraction to decompose this rational function

$$\rightarrow \frac{A}{1-2x} + \frac{B}{1-3x} + \frac{C}{1-x} + \frac{D}{(1-x)^2}$$

$$= A(1-3x)(1-x)^2 + B(1-2x)(1-x)^2 + C(1-2x)(1-3x)(1-x) + D(1-2x)(1-3x)$$

$$= A(1-3x)(1-2x+x^2) + B(1-2x)(1-2x+x^2) + C(1-2x)(1-3x)(1-x) + D(1-2x)(1-3x)$$

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$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 3 & 2 & 11 & 6 & 20 \\ 1 & -4 & -6 & -3 & -22 \\ 1 & 1 & 1 & 0 & 5 \end{array}\right) \xrightarrow{R_2 \to R_2 - 3R_1,\ R_3 \to R_3 - R_1,\ R_4 \to R_4 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & 8 & 3 & 2 \\ 0 & -5 & -7 & -4 & -28 \\ 0 & 0 & -1 & 1 & -1 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - 5R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & 1 & -8 & -3 & -2 \\ 0 & 0 & 33 & 11 & -18 \\ 0 & 0 & -1 & 1 & -1 \end{array}\right) \xrightarrow{R_4 \to 33R_4 + R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & 1 & -8 & -3 & -2 \\ 0 & 0 & 33 & 11 & -18 \\ 0 & 0 & 0 & 44 & -36 \end{array}\right)$$

$$\text{rank } A = 4 = \text{rank } [A:B]$$

$$\therefore \text{Consistent} \quad \text{and} \quad \rho(A) = 4 = \text{no. of unknowns}$$

$$\therefore \text{Unique solution}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 3 & 2 & 11 & 6 & 20 \\ 1 & -4 & -6 & -3 & -22 \\ 3 & -2 & -6 & 0 & -2 \end{array}\right) \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = \begin{pmatrix} 6 \\ 20 \\ -22 \\ -2 \end{pmatrix}$$

using R_2

$$\rightarrow A - 2B - 6C + 0D = 2 - 2(0) = 2$$

$$A(1 - 2x + x^2 - 3x(1 - 2x + x^2)) \quad A(1 - 5x + 9x^2 - 3x^3)$$

$$+ C(1 - 6x + 11x^2 - 6x^3) + D(1 - 12x + 5x^2 - 2x^3)$$

$$= a_0 + (2 - a_1)x + (5a_0 - a_1)x^2 + (7 - 2a_0)x^3$$

$$A + B + C + D = a_0 = 6$$

$$-5A - a_1 B - 6C - 5D = 2 - a_1(6) = 2 - 2a_1 = -22$$

$$3A + 2B + 11C + 6D = 5a_0 - a_1 = 5(6) - a_1 = 20$$

$$\rightarrow A

#2

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & 0 & 8 \\ 0 & 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 2 & 5 \end{array}\right)$$

$$2D = 5 \rightarrow D = 5/2 = D$$

$$C - \frac{1}{2}D = 0$$

$$C - \frac{1}{2}D = \frac{1}{2}(5/2) = 5/4 = C$$

$$B - C = 8 \rightarrow B = 8 + C = 8 + 5/4$$

$$= 32 + 5 = \frac{37}{4} = B$$

$$A + B + C + D = 6$$

$$A = 6 - 37/4 - 5/4 - 5/2 =$$

$$= 6 - \frac{42}{4} - 5/2 = 6 - \frac{42 - 10}{4}$$

$$= 6 - \frac{32}{4} = 6 - 13 = -\frac{7}{1} = A$$

$$= \sum_{n=0} A(2)^n + B(3)^n + C + D(n+1)$$

$$= -7(2)^n + \frac{37}{4}(3)^n + \frac{1}{2}(n+1) + 5/4$$

#2 $r^3 = 6r^2 - 12r + 8$

$\rightarrow r = 2, 2, 2$

$\rightarrow a_n = (A + Bn + Cn^2) r^n$

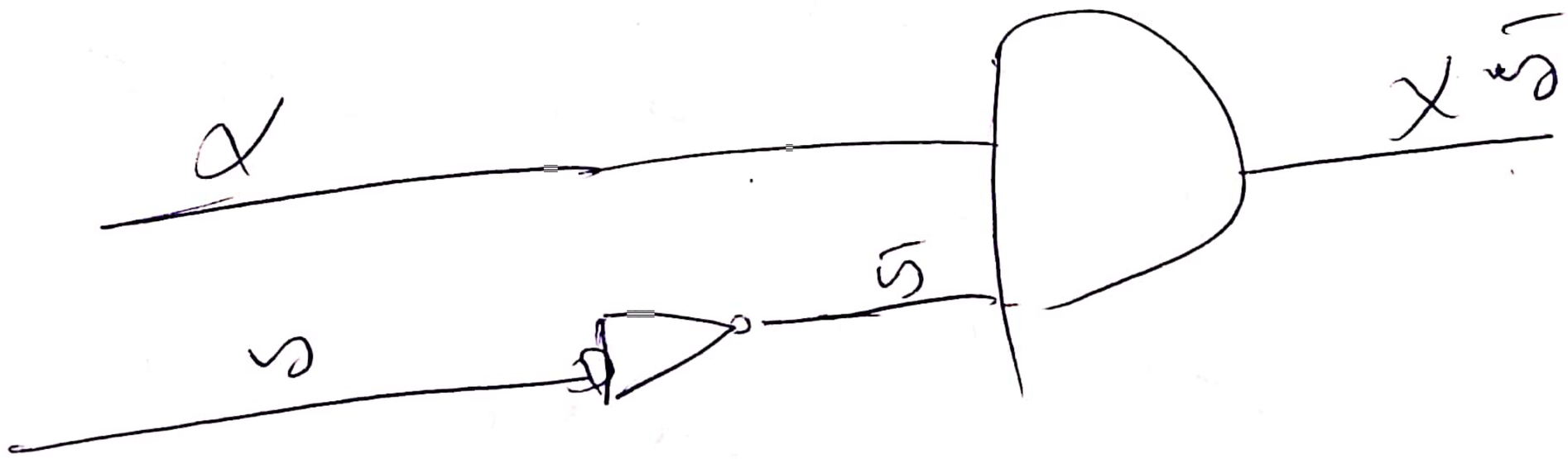
$a_n = (A + Bn + Cn^2)(2)^n$

$a_0 = 3,\ a_1 = 4,\ a_3 = 28$

$A = 3,\ B = -4,\ C = 3$

$\rightarrow a_n = (3 + -4n + 3n^2)(2)^n$

#3 $f(x,y,z) = x\bar{y}$ - Simplified



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Ch-4: Boolean Algebra

- Algebra - Studies about operations
  - Boolean Algebra about operation on 0,1
  - Study of mathematics
  - Boolean Algebra is Study of mathematics that provides and studies the operation and the rule for working with the set {0,1}
  - Here are some operation
    1) Complement $(\bar{x} \ni x \in \{0,1\})$ $\bar{1}=0, \bar{0}=1$
    2) Boolean Sum
       $1+1=1, \quad 1+0=1, \quad 0+0=0$
    3) Boolean product
       $1.1=1, \quad 1.0=0, \quad 0.1=0, \quad 0.0=0$

Example: Find the Value of $1+0+(\bar{0}+1)$

$= 1.0+(\bar{1}) = 1.0+0 = 0$

Boolean Expression and Boolean function

let $B=\{0,1\}$

- The variable x is called a Boolean variable if it assume values from B only.
- A function that involves Boolean variables is called Boolean function

$F: B^n \longrightarrow B$, $B^n = \{(x_1, x_2, \ldots, x_n)$ such that $x_i \in B = \{0,1\}, 1 \le i \le n$

Example: $f(x,y,z) = xy + \bar{z}$ is BF

i.e $F: B^3 \to B$, $x,y,z \in B$.

Boolean Expression

$E_1, (E_1 . E_2)$ and $(E_1 + E_2)$

Example: Find the values of the Boolean function representing $f(x,y,z) = xy + \bar{z}$

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$2^3 = 8$ (row)

x	y	z	xy	$\bar{z}$	$F(x,y,z) = xy + \bar{z}$
1	1	1	1	0	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	0	1	1
0	1	1	0	0	0
0	1	0	0	1	1
0	0	1	0	0	0
0	0	0	0	1	1

Equivalent Boolean Expression

Two different Boolean expression that represent the same out put (Value) are called equivalent.

Example

1) xy, $xy+0$, $xy.1$ are different expression but equivalent.

2) $xy+0 \neq xy+1 \neq xy$

Equivalent Boolean identities

- $\bar{\bar{x}} = x$ - double complement
- $x+x=x$ - Idempotent
  $x.x=x$ -
- $x+0=x$ } - identity
  $x.1=x$ }
- $x+1=1$ } - dominance
  $x.0=0$ }
- $x+y=y+x$ } - Commutative
  $xy=yx$ }
- $x+(y+z)=(x+y)+z$ } - Associative law
  $(xy)z=x(yz)$ }
- $\overline{xy} = \bar{x}+\bar{y}$, $\overline{(x+y)} = \bar{x}\bar{y}$ - Demorgan

(2) $x(y+z) = xy + xz$ - distributive law.

Example: Show using table

1) $x(y+z) = xy + xz$
2) $\overline{xy} = \bar{x} + \bar{y}$

Show that 1. $x + yz = (x+y)(x+z)$

2) $x(x+y) = x$

Soln (2): $x \cdot (x+y) = (x+0)(x+y)$

$= x + (xy + 0 \cdot x + 0 \cdot y$

$= x + xy = x(1+y) = x(1) = x$

Ex: Show the equivalence of the following

1 $xyz + \bar{x}y + xy\bar{z} = y$

2 $(x \cdot y)(\bar{x} + z)(y+z) = (x+y) \cdot (\bar{x} + z)$

3 $xy + \bar{x}z + yz = xy + \bar{x} \cdot z$

Logic gates

Boolean Algebra is used to model the Circuit of electronic devices like Computers.

Computer or other electronic devices is made up of number of circuits.

The basic elements of Circuit are gates

Using gates we will apply the Rules of Boolean algebra to design (Model) Circuits that performs a variety of tasks.

Here are some gates & their tasks

Inverter: x → ▷○ → $\bar{x}$ (Symbol)

Symbol

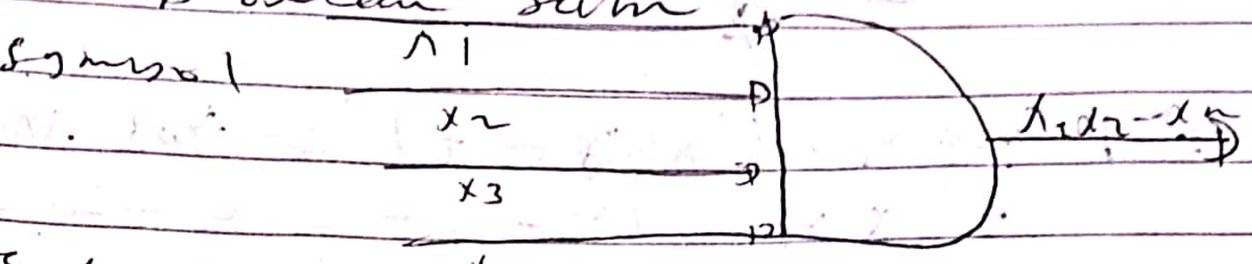
which accepts the value of one Boolean variable as input and produces the complement of this value as its output

OR Gate

Symbol

The inputs to this gate are Values of two or more boolean variables and the outputs is the Boolean sum.

AND gate : Symbol



The inputs to this gate are the values of two or more Boolean Variable; the output is the Boolean product of their values

Combination circuits

Can be constructed using the combination of inverter, OR gates and AND gates

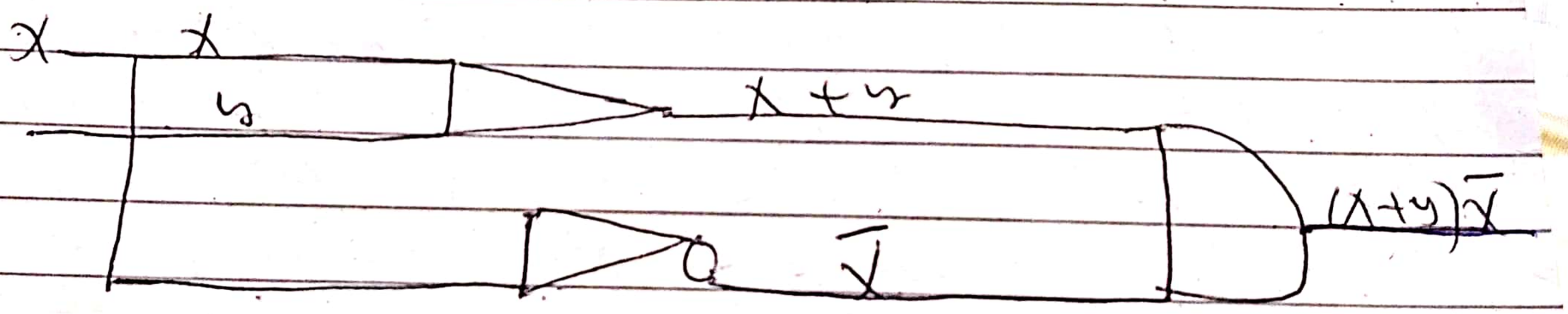
Example: Construct circuits that produces the following out puts.

a) $(x+y)\bar{x}$ b) $\bar{x}(y+z)$

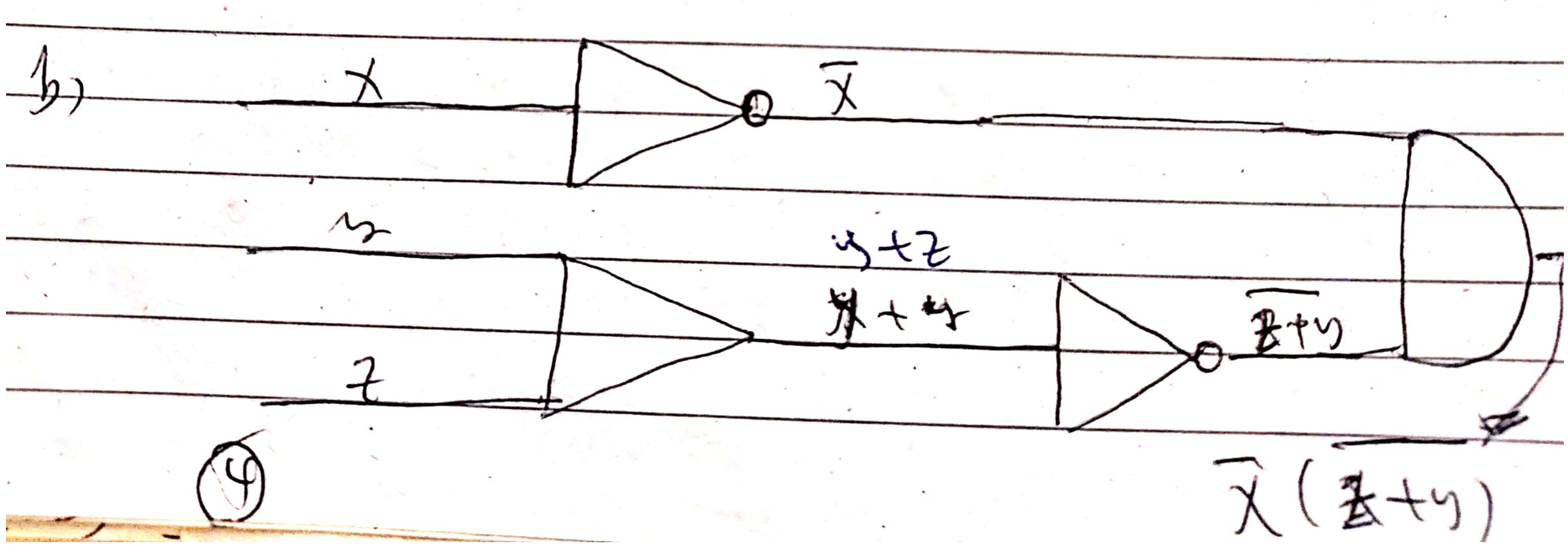
c) $(x+y+z)(\bar{x}\bar{y}\bar{z})$, d) $x\overline{(y+z)}$ — Exercise

Soln

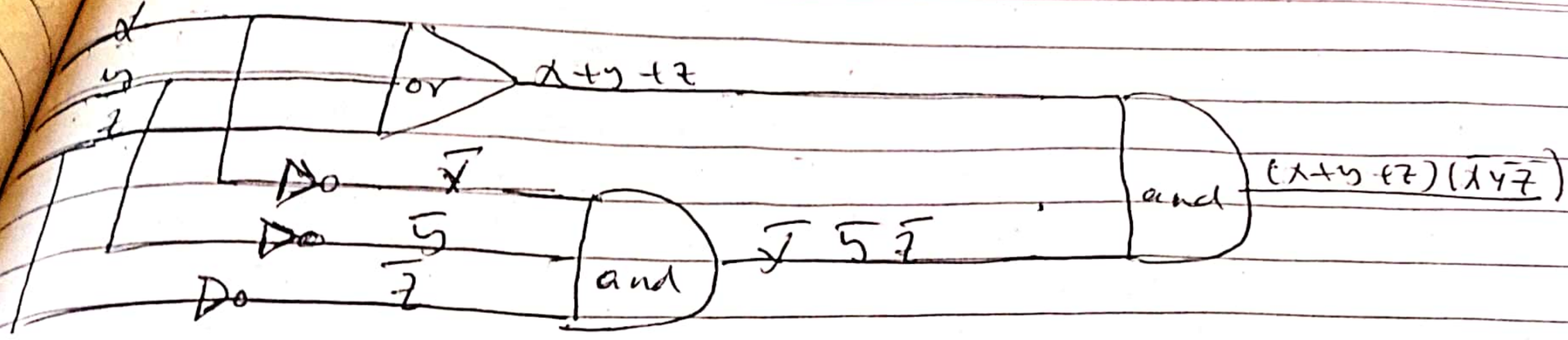
a)



b)



(4)



Minimization of Circuits

The efficiency and time of Combinational Circuits depends on the number of gates

2 To increases the efficiency and ~~time~~ minimize time we can minimize number of gates that is called minimization of circuits

Simplify to equivalent Boolean expression and draw the circuits

Example: simplify and draw Combination circuits

1) $xyz + x\bar{y}z$

2) $x + \underline{xy} + x\bar{y}z$

3) $xy + x(y+z) + y(y+z)$

4) $\underline{x\bar{y} + xy} + yz$

Sol:-

1) $xz(y+\bar{y}) = xz$

2) $x(1+y+\bar{y}+z) = x$

3)

$x + xy + x\bar{y}z$

$x(1+y) + x\bar{y}z$

$x.1 + x\bar{y}z$

$x + x\bar{y}z$

$= x(1+\bar{y}z)$

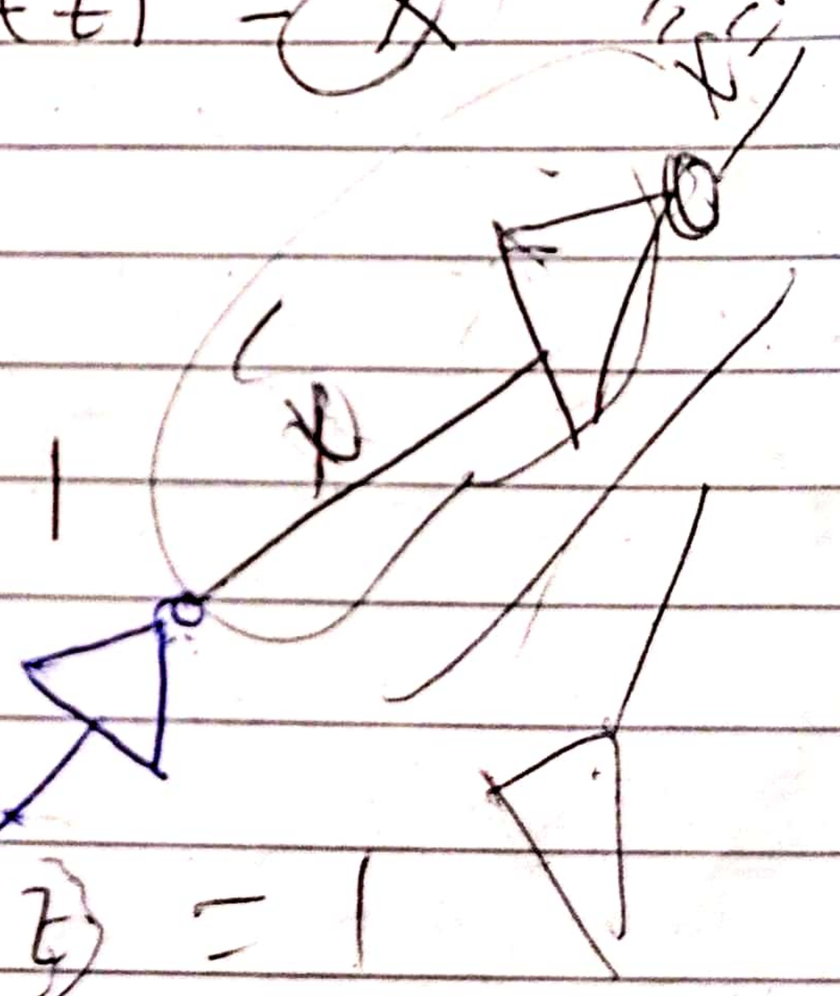
$= x.1$

$= x$

$x(1+y) + x\bar{y}z$

$1 + y = 1$

$1 + \bar{y} = 1$



$1 + yz = 1$

$$z \times (x+h) = z x + (1+x) h =$$

$$h \quad z x + (h + h x) =$$

$$h \quad D \; z x + h x =$$

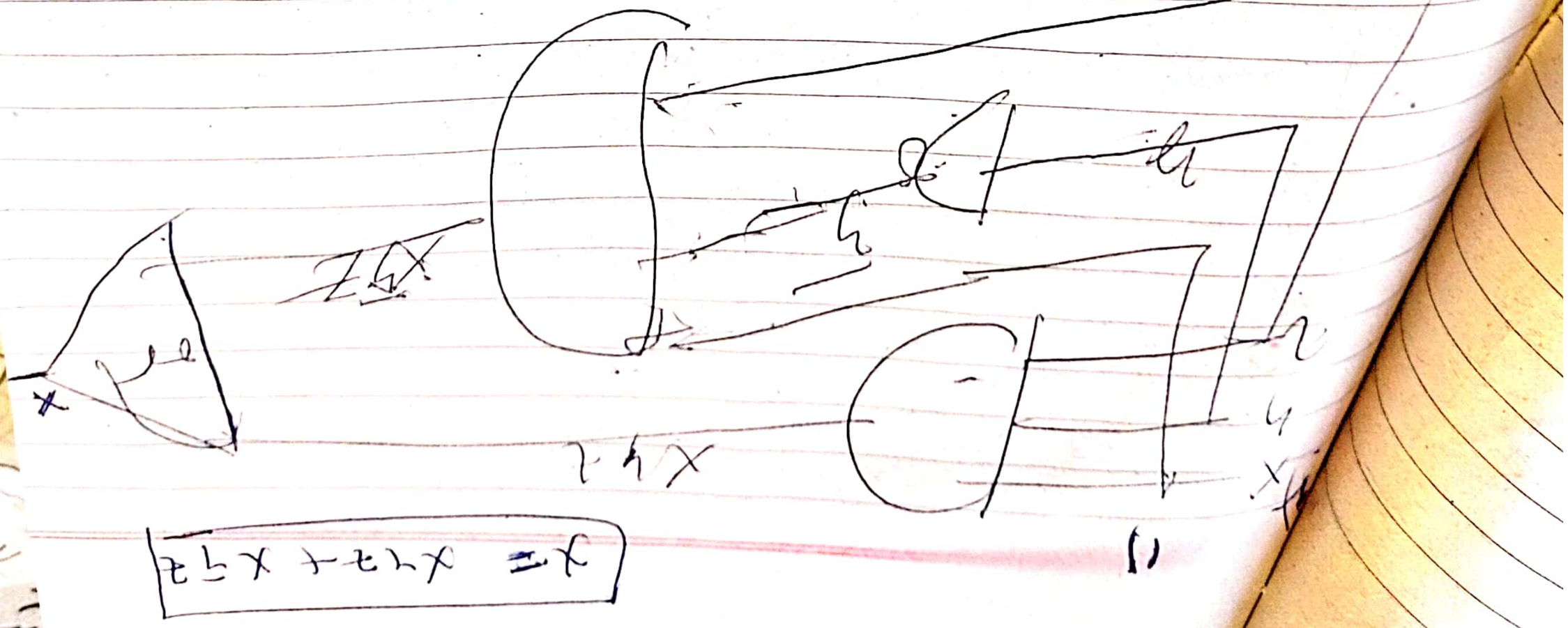
$$\underline{(z+1) h} + z x + h x =$$

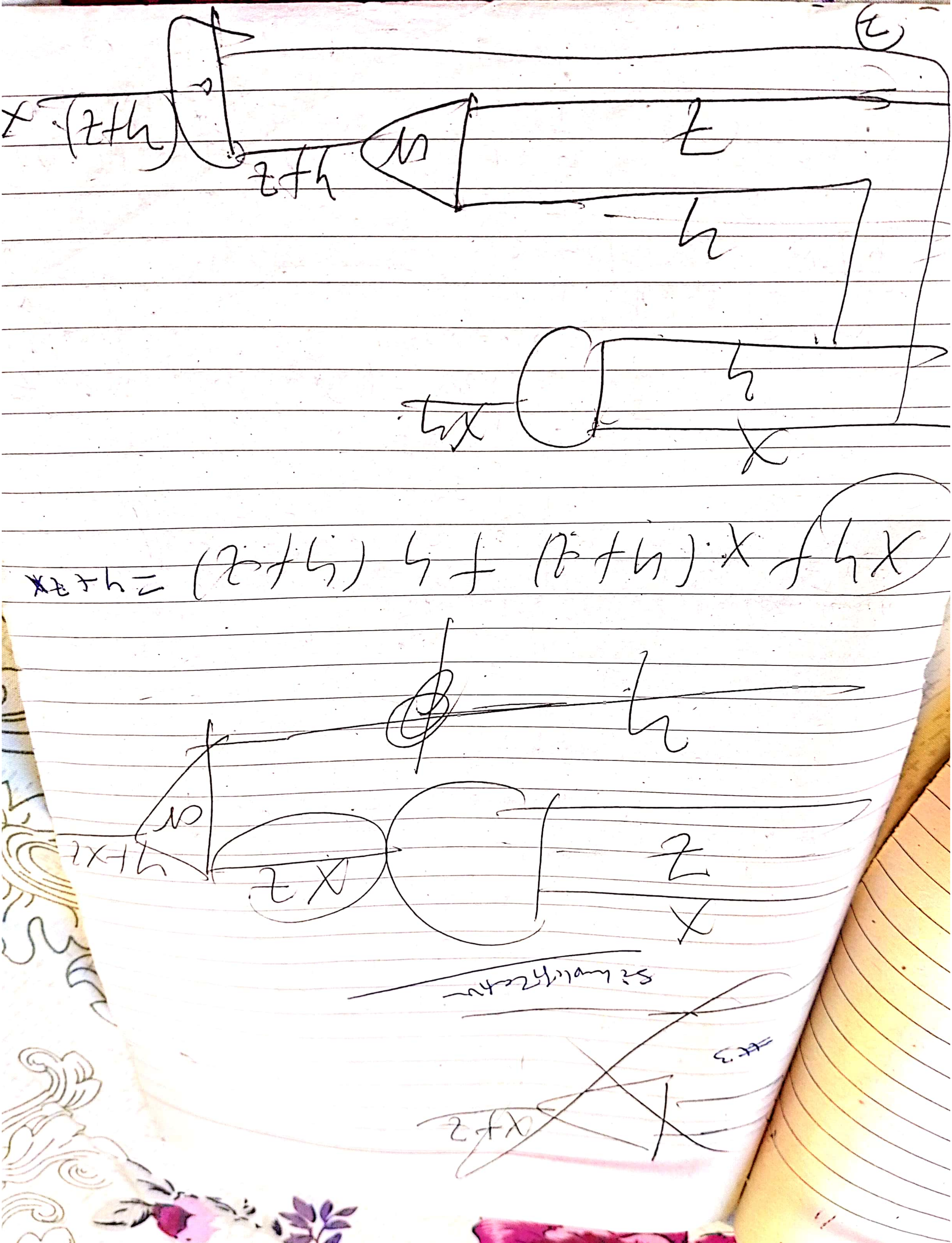
$$z h + h + z x + h x =$$

$$z h + h \; / \; z x + h x + h x =$$

$$(z+h) h + (z+h) x + h x$$

$$z x + h x + h x$$





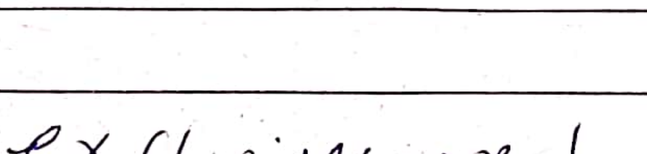
Values and Variables can indicate some of the following binary pairs
on/OFF, true/False, High/low, Closed/Open, 1/0

Logic Gates

1) Not (inverter) 2) OR 3) AND 4) NAND
5) NOR 6) Exclusive OR (XOR)
7) XNOR (Exclusive Nor)

4) Not AND (NAND) Gates

Symbol



truth table

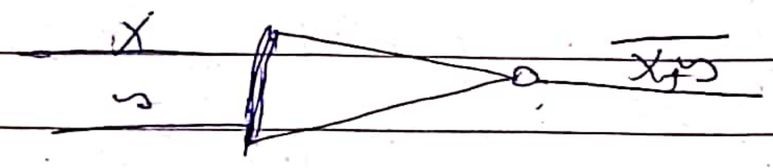
X	Y	F(X,Y)
0	0	1
0	1	1
1	0	1
1	1	0

Boolean

$F = \overline{XY} = \overline{X} + \overline{Y}$

5) Not OR (NOR gates)

Symbol



truth table

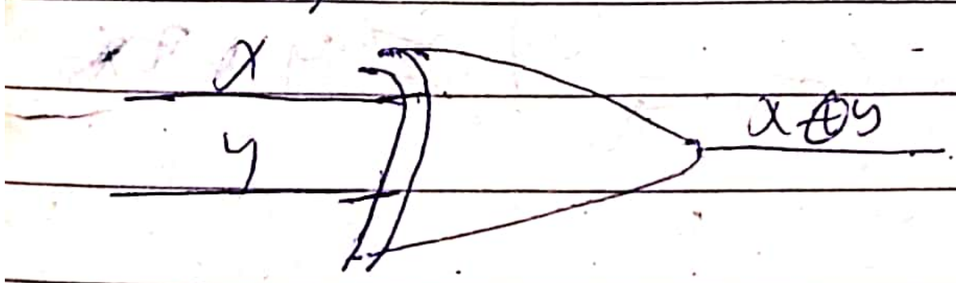
X	Y	f(X,Y)
0	0	1
0	1	0
1	0	0
1	1	0

Boolean

$F(X,Y) = \overline{(X+Y)}$

6) Exclusive OR (XOR gates)

Symbol



truth table

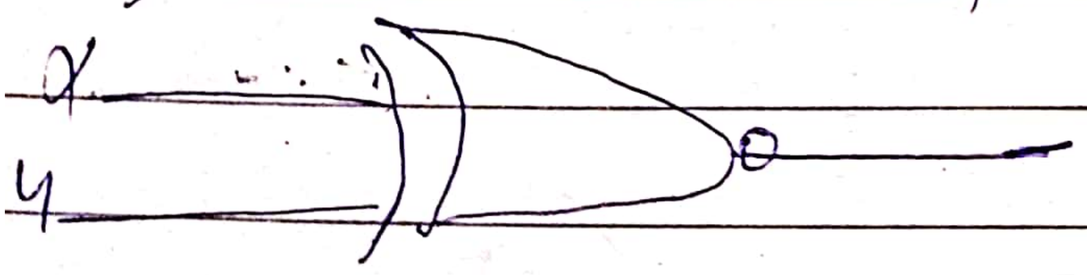
X	Y	f(X,Y)
0	0	
0	1	
1	0	
1	1	

Boolean function

$F(X,Y) = X \oplus Y$

7. Ex(Exclusive NOR) = XNOR

Symbol



truth table

X	Y	f(X,Y)
0	0	
0	1	
1	0	
1	1	

Boolean

$F(X,Y) = \overline{(X \oplus Y)}$

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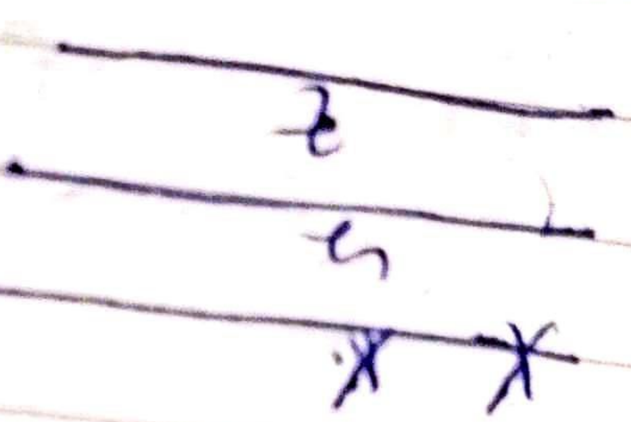

$$= x(1) + yz = x + yz,$$

$$= x + xy + yz = x(y+z) + yz$$

$$= xy + xz + yz$$

$$= xy + z(x+y) = xy + xz + yz$$

$$= (x+1)y + xz = y + xz$$

$$= x(y+z) + yz = xy + xz + yz$$

c) $x(y+z) + yz$

$$= xy + xz + yz$$

$$= xy + z(x+y) = xy + z(1+y)$$

$$= x + z$$

b) $x + xy + xz$

$$= x(1+y+z) = x(1+z)$$

$$= x$$